Grade 10 Chemistry: Chemical Bonding and Molecular Structure

1. Introduction to Chemical Bonding
Chemical bonding refers to the attractive forces that hold atoms or ions together to form stable molecules or ionic compounds.

Key Concepts
- Valence Electrons: Electrons present in the outermost shell of an atom involved in bond formation.
- Octet Rule: Atoms tend to gain, lose, or share electrons to attain eight electrons in their outer shell.
- Electronegativity: The tendency of an atom in a molecule to attract shared pairs of electrons toward itself.

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2. Types of Chemical Bonds

A. Ionic (Electrovalent) Bonding
Formed by complete transfer of one or more electrons from a metallic atom to a non-metallic atom.
- Example: Formation of Sodium Chloride (NaCl)
  $\text{Na} \rightarrow \text{Na}^+ + e^-$
  $\text{Cl} + e^- \rightarrow \text{Cl}^-$
  $\text{Na}^+ + \text{Cl}^- \rightarrow \text{NaCl}$

B. Covalent Bonding
Formed by mutual sharing of electron pairs between two non-metallic atoms.
- Single Covalent Bond: Shared pair of 1 electron pair (e.g., H_2).
- Double Covalent Bond: Shared pair of 2 electron pairs (e.g., O_2).
- Triple Covalent Bond: Shared pair of 3 electron pairs (e.g., N_2).

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3. Comparison of Ionic and Covalent Compounds

Property	Ionic Compounds	Covalent Compounds
Physical State	Hard crystalline solids	Gases, liquids, or soft solids
Melting/Boiling Points	High	Low to moderate
Electrical Conductivity	Conducts in molten/aqueous state	Non-conductors
Solubility	Soluble in water, insoluble in organic solvents	Soluble in organic solvents

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